Forest coordinates and relative generation

Human Compression project

Unified five-application draft

Abstract

We develop common generation, subgroup and flag arguments for five families of incidence geometries. A laminar extension theorem preserves a prescribed local map, so ordinary tree flags, parity conditions and four-point decorations use the same complete extension proof. Normal-quotient normalization and binary-module bounds give a single extremal symmetric-group classification used by two applications. We also retain complete primitive and imprimitive bounds, finite-core intersections and exact tree presentations needed by the high-rank converse. The local Frattini and graph-automorphism arguments remain in their applications. All source hypotheses and retained outside results are explicit; this edition makes no claim about its measured length before independent review.

1 Introduction

There are two different directions in these results. Once a tree or a finite decorated core is known, subgroup intersections and completion of flags can be proved by common constructions. Proving that every allowed marking has that form is a separate classification problem. We keep those directions separate and give complete proofs of both where the applications require them.

The common coset and relative-generation interfaces serve the original four applications. The laminar extension theorem strengthens their forest proof by retaining a prescribed map on a root set. The three four-point decorations then require only their actual local verification. The ordinary/sign extremal classification is proved once here, with Cameron–Cara and Whiston attribution, and supplies the involution-only maximum-rank classification as a corollary. The all-degree transitive parabolic exclusion remains a substantial additional argument; it is currently used by the high-rank application alone and is included in full.

A C-group marking is a finite generating family of distinct nonidentity involutions satisfying the intersection property for all subsets. No string commutations are assumed. Independence bounds below concern arbitrary group elements and do not require involutions or the intersection property. Finite-group and finite-degree assumptions are stated where needed.

2 Flags, parabolics and coset coordinates

All type sets are finite. Groups need not be finite unless this is stated. A typed incidence system has a reflexive symmetric incidence relation, with distinct elements of the same type never incident. A flag is a pairwise incident set. A chamber has every type. A geometry is such a system in which every flag extends to a chamber. It is firm if every nonchamber flag has at least two chamber extensions. A residue consists of the elements incident with a fixed flag and having a type absent from that flag. Residual connectedness means that every residue of rank at least two has a connected incidence graph. The specified group action is part of the data.

For subgroups $H_i \le G$, indexed by I, write

$$H_J = \bigcap_{j \in J} H_j, \qquad H_\varnothing = G, \qquad B = H_I.$$

The elements of type i are the left cosets of H_i, and two cosets of different types are incident if they intersect. The action is left multiplication. Inversion converts this convention to the right cosets of the sources. We will use the following three conditions:

(G1) The indexed intersections H_J are all distinct.

(G2) If $|I \setminus J| \geq 2$, then $H_J = \langle H_{J\cup\{k\}} : k \in I \setminus J \rangle$.

(G3) Every family of cosets of different types with pairwise nonempty intersections has a common element.

Proposition UB.Coset (the complete coset interface). Conditions (G1)–(G3) hold if and only if the coset system is firm, residually connected and flag-transitive under its given action. The residue at the standard flag of type J is the coset geometry of H_J with subgroups $H_{J\cup\{k\}}$, $k \notin J$.

Proof. Under (G3), a flag has a representative g common to its cosets and is therefore contained in the chamber $(gH_i : i \in I)$. Translating g to the identity shows flag transitivity. Elements of type k incident with the standard flag of type J can, again by (G3), be represented by an element of H_J. Two such representatives give the same coset precisely when they differ by $H_J \cap H_k$. This proves the residue description. Condition (G1) gives $H_{I\setminus\{i\}} > B$, so every panel has at least two elements. Every shorter flag is contained in a panel; this proves firmness.

For completeness, a coset incidence graph for subgroups $(L_i : i \in S)$ of a group L, with $|S| \geq 2$, is connected exactly when the L_i generate L. Every edge can be represented by a common group element. Successive representatives along a path differ by elements of the subgroup corresponding to their shared vertex. Thus a connected path from the identity cosets to gL_i places g in $\langle L_j : j \in S \rangle$. Conversely, for $a \in L_j$ and $i \neq j$, the vertices $L_i, L_j = aL_j, aL_i$ give a path. Translating and concatenating these paths for a word in the L_j connects every coset to the standard ones; the standard vertices are mutually incident. Applying this criterion to the described residues proves residual connectedness from (G2).

Conversely, flag transitivity takes any pairwise incident family to the standard flag of its types, so it has a common representative and (G3) holds. Firmness and panel transitivity give an element $s_i \in H_{I\setminus\{i\}} \setminus B$ for every i. Its intersection with H_i is B, hence $s_i \notin H_i$. If $J \neq K$, choosing a type in one set difference shows that s_i belongs to one of H_J, H_K and not to the other. This proves (G1). Residual connectedness, the residue description and the graph criterion just proved give (G2). □

An action is primitive when it is transitive and has no nontrivial invariant partition. A geometry with its group action is PRI if the action on every type is primitive, and WPRI if the action on at least one type is primitive. The residual versions, RPRI and RWPRI, apply to each residue of positive rank, including the whole geometry. In coset coordinates, RWPRI says

$$J \subsetneq I \quad \Longrightarrow \quad \text{some } H_{J\cup\{k\}} \text{ is maximal in } H_J, \quad k \notin J. \tag{G4}$$

This is the elementary equivalence between primitivity on a coset space and maximality of its stabilizer. It refers to the stabilizer's induced action; a kernel of that action does not change its invariant partitions.

Lemma UB.Flags (residual weak primitivity gives flags). A finite-rank geometry with a residually weakly primitive group action is flag-transitive.

Proof. Fix a chamber. In the empty-flag residue choose a type on which the group is primitive, hence transitive. Match the element of that type in any second chamber to the fixed one. In the residue of the matched flag, choose a remaining primitive type and use its flag stabilizer to match the next element. At most $|I|$ steps match the chambers. Since every flag extends to a chamber, transitivity follows for flags of every fixed type. The typewise bijection from cosets of stabilizers to elements preserves incidence: transitivity on pairs of incident elements supplies a common representative. □

For $B \le G$, a finite indexed family $(s_i : i \in I)$ is independent relative to B if $s_i \notin \langle B, s_j : j \neq i\rangle$ for every i. If it also generates G together with B, it is a relative generating family. For finite G let $\mu(G,B)$ and $\mu'(G,B)$ be the largest cardinalities of relative independent generating families and relative independent families, respectively. Omit B if $B = 1$. Independence for a group acting on another group means independence under generation by all images of the other family members; it is denoted $\mu'_G(N)$ in the normal-quotient application.

3 Relative generation from residual maximality

For a finite indexed subgroup family $H_i \le G$, write $H_J = \bigcap_{j\in J} H_j$, with $H_\varnothing = G$ and $B = H_I$.

Theorem UB.Relative (generation from residual maximality). Suppose all H_J, $J \subseteq I$, are distinct, and for every $J \subsetneq I$ there is $k \notin J$ such that $H_{J\cup\{k\}}$ is maximal in H_J. Choose

$$s_i \in H_{I\setminus\{i\}} \setminus H_i. \tag{1}$$

Then

$$H_J = \langle B, s_i : i \notin J\rangle \quad (J \subseteq I). \tag{2}$$

In particular these elements generate G irredundantly relative to B, and their relative parabolics satisfy

$$\langle B, s_i : i \in S\rangle \cap \langle B, s_i : i \in T\rangle = \langle B, s_i : i \in S \cap T\rangle. \tag{3}$$

The group need not be finite.

Proof. Distinctness of $H_{I\setminus\{i\}}$ and B makes (1) possible, since its intersection with H_i is B. Every s_j with $j \neq i$, and every element of B, belongs to H_i, while s_i does not. This proves relative independence.

Prove (2) by downward induction from $J = I$. Choose k as in the hypothesis. By induction, $H_{J\cup\{k\}}$ is generated by B and the s_i with $i \notin J \cup \{k\}$. The element s_k belongs to H_J, because $J \subseteq I \setminus \{k\}$, and lies outside $H_{J\cup\{k\}}$, because it lies outside H_k. Maximality therefore says that adjoining s_k generates H_J. This is (2). Finally substitute $J = I \setminus S$ and $I \setminus T$ into the elementary equality $H_J \cap H_{J'} = H_{J\cup J'}$ to obtain (3). □

Theorem UB.Generation (the geometric interface). For a finite-rank firm geometry with a residually weakly primitive action, choose one chamber and its type stabilizers H_i. Every choice

$$s_i \in H_{I\setminus\{i\}} \setminus H_i$$

is independent relative to $B = H_I$, generates G over B, and satisfies

$$H_J = \langle B, s_i : i \notin J\rangle \qquad (J \subseteq I).$$

Every such geometry is residually connected. For finite G its rank is at most $\mu(G,B)$. Without residual weak primitivity, a firm flag-transitive geometry still has a relatively independent family of panel choices and, for finite G, rank at most $\mu'(G,B)$.

Proof. Lemma UB.Flags supplies flag transitivity and coset coordinates for the first assertion. Firmness gives distinct indexed intersections by the panel argument in Proposition UB.Coset. Residual weak primitivity is exactly (G4), so Theorem UB.Relative applies to the displayed choices. Its formula implies (G2): if at least two types remain, each of the remaining generators belongs to an intersection obtained by fixing another remaining type. Consequently Proposition UB.Coset gives residual connectedness. The relative generating family has size equal to the rank, proving the bound by $\mu(G,B)$.

For the last assertion no maximality is needed. Firmness and flag transitivity again give panel choices. For a fixed i, all other choices and B lie in H_i, whereas s_i does not. They are therefore relatively independent, which gives the bound by $\mu'(G,B)$. □

4 Extending prescribed maps through laminar flags

A family of subsets is laminar if any two are nested or disjoint. The following statement retains an arbitrary prescribed local map; that is the part not stated in the original UB.F interface.

Theorem 1 (UB.Laminar)**.** *Let Ω be finite, let $Q \subseteq \Omega$, and let $(A_i)_{i\in I}$ be a finite indexed laminar family of subsets of $\Omega \setminus Q$. Choose $J \subseteq I$ and subsets X_i, $i \in J$, with*

$$|X_i| = |A_i|, \qquad |X_i \cap X_j| = |A_i \cap A_j|.$$

Write $D = \Omega \setminus \bigcup_{i\in J} X_i$. Every injection $u : Q \to D$ extends to a permutation g of Ω satisfying $gA_i = X_i$ for all $i \in J$.

Proof. If a selected set is Ω, then $Q = \varnothing$ and its target is necessarily Ω. Discard that vacuous constraint; the prescribed map is empty in this case. We may therefore assume that every selected set is proper, using the complement of the remaining targets as the root region. The intersection cardinalities enforce the same disjointness and containment relations among the X_i as among the A_i. Equal standard sets give equal target sets and can be treated once. Adjoin Ω above all the selected sets. For each selected container, subtract its maximal selected proper subcontainers. Since those subcontainers are disjoint, the difference has size equal to the container's size minus the sum of their sizes. The corresponding standard and target differences therefore have equal cardinalities. They partition Ω on each side. The root difference contains Q on the standard side and is exactly D on the target side. Extend u to a bijection of these two differences, and choose arbitrary bijections on every other corresponding pair of differences. Their union is the required permutation. □

Lemma 2 (UB.Character)**.** *Suppose at least one permutation realizes prescribed laminar images and a prescribed partial bijection. Fix one such permutation g_0, and let K be the subgroup preserving every standard selected subset and fixing every point in the domain of that partial bijection. All realizations are exactly g_0K. For a homomorphism $\chi : S_\Omega \to C$ and $c \in C$, a realization with $\chi(g) = c$ exists exactly when $\chi(g_0)^{-1}c \in \chi(K)$.*

Proof. Two realizations differ on the right by a permutation satisfying exactly the defining conditions of K, and every such right multiplication preserves the prescribed data. Applying χ proves the second claim. □

These statements make explicit the extension freedom in the familiar atom-matching proof. Their use below does not claim a new historical classification or a prior-art determination.

5 Weighted forests and complete flags

Let T be a finite tree with vertex set V and edge set E. Give each vertex a finite nonempty block B_v, with the blocks pairwise disjoint, and put $\Omega = \bigsqcup_v B_v$. For $F \subseteq E$, let

$$K_F = \prod_C S_{\bigcup_{v\in C} B_v}, \tag{4}$$

where C runs over the components of the forest (V, F). Root T, and let $A_e \subseteq \Omega$ be the union of blocks on the side of $T - e$ away from the root. In the finite symmetric group S_Ω, set $H_e = K_{E\setminus\{e\}} = S_{A_e} \times S_{\Omega\setminus A_e}$.

Theorem UB.F (forest intersections and flag completion). The following hold.

1. $K_F \cap K_D = K_{F\cap D}$. Moreover K_F is generated by $K_\varnothing$ together with the groups $K_{\{e\}}$, $e \in F$.
2. The coset system of the H_e is a firm, residually connected, flag-transitive geometry. Its type-e elements identify with the $|A_e|$-subsets of Ω, and distinct types satisfy

$$X_e * X_f \quad \Longleftrightarrow \quad |X_e \cap X_f| = |A_e \cap A_f|. \tag{5}$$

3. Its Borel subgroup is $K_\varnothing = \prod_v S_{B_v}$. The residue whose remaining types are F is the join of the same constructions on the nontrivial components of (V, F). Its acting group is K_F; components with no edges contribute only a kernel on this residue.

Proof. A permutation belongs to both groups in (1) exactly when it preserves every intersection of their component blocks. Two vertices lie in a common block of both forests precisely when their unique tree path belongs to both edge sets. Thus the common refinement is the component partition for $F \cap D$. Transpositions inside the blocks, together with transpositions joining adjacent blocks along retained edges, connect each component; they generate its full symmetric group. This proves (1).

The coset gH_e maps to gA_e. Two pairs of subsets are conjugate under S_Ω exactly when their four Boolean atoms have the same sizes, giving (5). The sets A_e are laminar: any two are nested or disjoint. Equation (5) forces the identical nesting and disjointness relations on any pairwise incident family (X_e). Theorem UB.Laminar, with no prescribed point map, gives a single permutation sending every selected A_e to X_e. This proves (G3) for every partial flag.

Repeated use of (1) gives $H_J = K_{E\setminus J}$. Different edge sets give different component partitions and different groups, since every block is nonempty and adding an edge allows a transposition between its two sides. Thus (G1) holds. If at least two edge types remain, the groups obtained by deleting each remaining edge collectively contain $K_\varnothing$ and each remaining one-edge group, proving (G2). The coset criterion gives the geometry assertions. Its residue group and its subgroups factor over the forest components; all elements from different components are incident, which gives the stated join and the kernel convention. ■

For unit blocks $B_v = \{v\}$, write τ_e for the endpoint transposition of e. The theorem says

$$K_F = \langle \tau_e : e \in F \rangle = \prod_C S_C, \qquad K_\varnothing = 1. \tag{6}$$

This is the unweighted model used for [BC98]. The weighted formulation also supplies the parity construction after an edge contraction and the disjoint-subset construction after a star specialization.

6 Overgroups of a two-block stabilizer

Lemma UB.Overgroups (two-block overgroups). Let $\Omega = A \sqcup C$ be a finite set, with $a = |A|$, $c = |C|$ positive, and put $H = S_A \times S_C$. If $a \neq c$, or $a = c = 1$, then H is maximal in S_Ω. If $a = c \geq 2$, its only proper strict overgroup is the unordered two-block stabilizer

$$W = (S_A \times S_C) \rtimes C_2,$$

which is maximal and satisfies $[W : H] = 2$. In all cases the subgroup interval $[H, S_\Omega]$ is a chain. A positive weighted tree has at most one edge whose two sides have equal total weight.

Proof. The case $a = c = 1$ is immediate. The group H contains the transpositions within each of its two blocks. If an overgroup element does not preserve their unordered partition, its conjugates of these transpositions include one crossing the original blocks. Together with the original within-block transpositions the support graph is connected, so these transpositions generate S_Ω. This remains valid when one block is a singleton: a permutation moving that singleton produces a cross-block conjugate from the other block. Thus a proper overgroup must preserve the unordered partition. That stabilizer equals H when the sizes differ and has index two over it when they agree. Its only further overgroup is S_Ω, by the same argument.

If an edge bisects the total weight, the side of any other edge away from it is a proper positive-weight subset of one of its halves. It has strictly less than half the total weight and cannot be a bisecting side. ■

7 Maximum irredundant length and a normal module

A generating family is irredundant if deleting any member leaves a proper subgroup. Write $\mu(G)$ for the largest size of an irredundant generating family of a finite group G, and $\mu'(G)$ for the largest size of any independent family in it. These are maximum lengths, not the minimum number of generators. For an abelian normal subgroup $V \trianglelefteq G$, let $\mu'_G(V)$ denote the largest size of a family independent under generation by all G-conjugates.

Lemma 3 (UB.NormalQuotient)**.** *For finite G and $N \trianglelefteq G$, $\mu(G) \le \mu(G/N) + \mu'(N)$. If N is abelian, the final term can be replaced by $\mu'_G(N)$. In particular, if $V \trianglelefteq G$ is an elementary abelian minimal normal subgroup, then*

$$\mu(G) \le \mu(G/V) + 1. \tag{7}$$

No splitting of the extension is needed.

Proof. Take an irredundant generating family S of G, and an inclusion-minimal subfamily T whose images generate G/N. For $s \in S \setminus T$, choose a word w_s in T with $sN = w_sN$, and put $v_s = sw_s^{-1} \in N$. If one v_s lies in the subgroup generated by the others, substitution expresses s using $S\setminus\{s\}$, a contradiction. Since $|T| \le \mu(G/N)$, the first bound follows.

When N is abelian, $G = N\langle T\rangle$ implies that its conjugation action on N is already the action of $\langle T\rangle$. Thus an expression for v_s in conjugates of the other v_t again substitutes to an expression for s without s. This proves the normal-module bound. If V is elementary abelian minimal normal, it is an irreducible $\mathbb{F}_pG$-module. Any nonzero vector generates all of V under conjugation, so an independent family for this action has at most one member. This proves (7) for maximum irredundant length. □

This is Cameron–Cara's normal-quotient theorem [CC, Theorem 1.1]. Both the extremal and the high-rank applications use this one complete proof. We retain Whiston's bound that an independent family in S_m has at most $m - 1$ members, with equality only when it generates S_m. In particular $\mu'(A_m) \le m - 2$. No classification of the C-groups of PSL_2 or PGL_2 is used below.

Corollary 4. *If G has a faithful primitive affine action with elementary abelian translation subgroup V, then* (7) *applies. In particular $\mu(\mathrm{AGL}(4,2)) \le 7$.*

Proof. The translation module is irreducible: a proper invariant subspace would give a nontrivial block system. Thus V is minimal normal. For $\mathrm{AGL}(4,2)$ the quotient is $\mathrm{GL}(4,2) \cong A_8$, so Whiston gives $\mu(A_8) \le 6$. The classical exceptional isomorphism is a retained identification; the finite certificate also checks an explicit isomorphism for the catalogue representation used below. □

8 All transitive imprimitive groups of degree nine

Theorem 5. *Every transitive subgroup of $S_3 \wr S_3$ in its imprimitive action on nine letters has $\mu(G) \le 5$. Consequently it cannot be a maximal parabolic of a rank-seven C-group marking of S_9.*

Proof. Let K be the kernel of the action on the three blocks, and put $A = K \cap A_3^3$. Then A is an elementary abelian normal 3-subgroup, viewed as an invariant subspace of $\mathbb{F}_3^3$. The quotient G/A has normal subgroup $W = K/A \le C_2^3$ and transitive quotient $B \le S_3$, so $B = C_3$ or S_3.

Choose an element of G inducing a three-cycle on the blocks. On A_3^3 its conjugation action is a cyclic permutation of the coordinates with coefficients $\varepsilon_i \in \{1, -1\}$. The vector in any one coordinate is cyclic for this operator, whose characteristic and minimal polynomials are $X^3 - \delta = (X - \delta)^3$, where $\delta = \varepsilon_1\varepsilon_2\varepsilon_3 \in \mathbb{F}_3$. Thus the full module is $\mathbb{F}_3[X]/(X - \delta)^3$. Its invariant subspaces form a chain, and each is cyclic. Any family of vectors in an invariant subspace therefore has one vector whose cyclic span contains all the others. Independence under the whole group implies independence under this one element, so $\mu'_G(A) \le 1$.

The corresponding action on C_2^3 is ordinary coordinate permutation. Under the three-cycle it is the direct sum of a trivial line and the irreducible plane with polynomial $X^2 + X + 1$. Every invariant subspace is a sum of a subset of these two nonisomorphic irreducibles. The two polynomial projection operators show that a vector generates each component on which it is nonzero. At most two vectors are therefore needed from any given family to generate its cyclic submodule; a family independent under the full group has at most two members. Hence $\mu'_{G/A}(W) \leq 2$.

Twice applying Lemma 3 gives

$$\mu(G) \leq \mu(G/A) + 1 \leq \mu(B) + 2 + 1 \leq 5.$$

This applies to the maximum irredundant length, not just a selected small generating set. A parabolic of rank six in a C-group has six independent distinguished generators, contradicting the bound. □

9 Independent vectors in a transitive binary module

The following elementary observation is useful beyond the present classification. It bounds irredundance of a particular family, rather than merely providing some short generating family for the module.

Lemma 6. *Let M be a finite-dimensional module over a finite field, generated irredundantly as a module by $v_1, \ldots, v_s$. There are simple quotients $L_1, \ldots, L_s$ of M and a surjection $M \longrightarrow \bigoplus_{i=1}^{s} L_i$.*

Proof. For each i, choose a maximal proper submodule M_i containing all v_j with $j \neq i$. It does not contain v_i, since those vectors generate M. Put $L_i = M/M_i$. Under the product quotient map, the submodule generated by v_i has zero coordinates except the ith, where its image is the nonzero submodule L_i. Thus the image contains every direct summand and is the whole direct sum. □

Lemma 7. *Let $P \leq S_m$ be transitive, where $m \geq 4$, acting by coordinate permutation on $V = \mathbb{F}_2^m$. For every invariant subspace $W \leq V$, every P-independent family in W has size at most $m - 2$. For $P = S_m$ the stronger bound two holds.*

Proof. Let M be the submodule generated by an independent family of size s. If $M < V$ and $s \geq m-1$, Lemma 6 forces $\dim M = s = m - 1$ and every simple quotient there has dimension one. Over $\mathbb{F}_2$ a one-dimensional representation is trivial, so the surjection is an isomorphism and M is fixed pointwise by P. But the fixed space of a transitive permutation group on V is the one-dimensional constant subspace, a contradiction.

If $M = V$, at most one of those simple quotients has dimension one: otherwise the product surjection would give two linearly independent maps from V to the trivial module, whereas a P-invariant linear functional has equal coefficients in all m positions. Hence $m \geq 2s-1$, and $s \leq \lfloor (m+1)/2 \rfloor \leq m-2$.

For the final assertion, if every vector in the family is constant, one suffices. Otherwise choose a nonconstant vector v. Subtracting its image under a transposition that exchanges a zero and a one gives $e_i + e_j$. Conjugation then generates the entire even-weight subspace. This subspace has codimension one, so at most one further member of the family is needed. Independence therefore permits at most two. □

Theorem 8. *Every transitive subgroup of $S_2 \wr S_m$ in degree $2m$, $m \geq 5$, has maximum irredundant generating length at most $2m - 4$.*

Proof. Let $W \leq \mathbb{F}_2^m$ be the block kernel and $P \leq S_m$ the transitive block image. The conjugation action on W is coordinate permutation. If $\mu(P) \leq m - 2$, Lemmas 3 and 7 give $\mu(G) \leq 2m - 4$. Otherwise Whiston's equality clause forces $P = S_m$. The sharper module bound gives $\mu(G) \leq (m - 1) + 2 = m + 1 \leq 2m - 4$. □

Lemma 9. *For $m \geq 4$, every transitive subgroup of $S_3 \wr S_m$ has $\mu(G) \leq 3m - 4$. For $m \geq 3$, every transitive subgroup of $S_4 \wr S_m$ has $\mu(G) \leq 4m - 4$.*

Proof. For blocks of size three, write K for the block kernel, $A = K \cap A_3^m$, $W = K/A \le \mathbb{F}_2^m$, and $P = G/K$. If $P \neq S_m$, then $\mu(P) \le m-2$, the binary-module cost is at most $m-2$, and the ternary-module cost is at most its ambient dimension m. This gives $3m-4$.

If $P = S_m$, the binary cost is at most two. The ternary cost is at most $m-1$. Indeed, an independent family of m vectors would span all of $\mathbb{F}_3^m$, and Lemma 6 would identify this monomial module with a sum of one-dimensional modules. Its action would then have abelian image. But its permutation image on the coordinate axes is S_m, which is nonabelian. Thus the total cost is at most $(m-1)+2+(m-1) = 2m \le 3m-4$. For three blocks of size three use Theorem 5 instead.

Now let the blocks have size four. The block-kernel projections are conjugate. If they are proper subgroups of S_4, every independent family in each projection has at most two members, by Whiston's equality clause. Successive projections of the kernel and Lemma 3 bound its independent families by $2m$. Consequently $\mu(G) \le (m-1) + 2m \le 4m-4$.

Otherwise every projection is S_4. Put $A = K \cap V_4^m$, where V_4 is the normal Klein four subgroup of S_4. Then G/A has a faithful transitive imprimitive action with m blocks of size three, so the preceding bound gives $\mu(G/A) \le 3m-4$. We claim that the normal-module cost of $A \le \mathbb{F}_2^{2m}$ is at most m. The group $L = K/A$ is a subdirect subgroup of S_3^m. Its normal subgroup $Q = L \cap C_3^m$ projects onto C_3 in every coordinate: the commutator subgroup of L already has this property. Thus Q fixes no nonzero vector in the full $2m$-dimensional module, since the natural C_3 action on each two-dimensional coordinate has no fixed vector. Averaging over the odd-order group Q shows that taking Q-fixed vectors is exact, so every subquotient of this module also has zero Q-fixed space. In particular every simple G-module quotient in Lemma 6 has dimension at least two. Hence a normal-independent family in A has size at most $\dim A/2 \le m$. The normal-module inequality now gives $\mu(G) \le (3m-4) + m = 4m-4$. □

10 A bounded primitive certificate

For an integer $a > 0$, let $\Omega(a)$ be its number of prime factors with multiplicity. Every strictly ascending subgroup chain in a finite group of order a has at most $\Omega(a)$ strict inclusions, because the product of its successive integral indices divides a. Thus the C-rank is at most $\Omega(|G|)$.

The required program `primitive_bound.g` enumerates the complete GAP primitive-group catalogue in degrees 9–25, omitting A_n, S_n, and reports exactly twelve groups for which this elementary bound exceeds $n-4$. All remaining catalogue entries are thereby discharged. The twelve exceptions are:

9	$3^2 : QD_{16}$, $3^2 : SL_2(3)$, $AGL_2(3)$, $PSL_2(8)$, $PSL_2(8) : 3$;
10	$A_6 : 2$, S_6, $A_6 2$, $A_6 : 2^2$;
11	M_{11};
12	M_{12};
16	$AGL_4(2)$.

(8)

The program verifies the precise subgroup and quotient identifications used next. Its retained inputs are the completeness of this catalogue and GAP's exact finite-group algorithms; it performs no search for C-group markings and imports none of the source's reported C-ranks.

For the first two degree-nine groups the irreducible translation module costs one and the quotient has at most four prime factors, giving five. For $AGL_2(3)$ the quotient $GL_2(3)$ has Frattini subgroup of order two and Frattini quotient S_4. Passing to a Frattini quotient preserves irredundant generating families: a subset generates the whole group exactly when its image generates the quotient. Thus $\mu(GL_2(3)) = \mu(S_4) \le 3$, and the affine group has $\mu \le 4$. Every maximal subgroup of $PSL_2(8)$ has order 56, 18 or 14; these have at most four prime factors, so a C-group marking has rank at most five. The extension by 3 is not generated by its involutions: the program checks that they generate only the index-three subgroup.

In degree ten the three groups other than S_6 have normal socle A_6 and quotient of order two or four. Lemma 3 and $\mu'(A_6) \le 4$ give at most six generators; S_6 has at most five. Corollary 4 treats degree sixteen.

For M_{11} the maximal-subgroup orders are $720, 660, 144, 120, 48$. The subgroup generated by all involutions in each has at most six prime factors; in the order-720 case its order is 360. Any C-group parabolic lying in such a maximal subgroup lies in its involution-generated subgroup. Hence M_{11} has C-rank at most seven. For M_{12} the maximal-subgroup orders are

$$7920, 7920, 1440, 1440, 660, 432, 432, 240, 192, 192, 72.$$

The first two are M_{11}. The order-1440 groups have normal A_6 with quotient order four, so their irredundant lengths are at most six. Every remaining maximal subgroup has at most seven prime factors. Deleting one distinguished generator and placing the resulting proper parabolic in a maximal subgroup gives C-rank at most eight for M_{12}. All these finite assertions are checked by the complete program.

For arbitrary degree $n \ge 26$, retain Maróti's Theorem 1.1 in its original form, including the lower containment $A_m^\ell \le G$ in the product-action alternative. If $\ell \ge 2$, the same abstract group acts faithfully on $m\ell$ points, so Whiston bounds its independent families by $m\ell - 1 \le m^\ell - 4 \le n - 4$; the inequality holds for $m \ge 2$, $\ell \ge 2$, $m^\ell \ge 9$. If $\ell = 1$, the natural action case is excluded by $G \ne A_n, S_n$; otherwise the k-subset action has $k \ge 2$ and $m \le n/2$, so the same conclusion follows from $m - 1 \le n - 4$. The Mathieu alternative has degree at most 24.

In the remaining alternative, Maróti bounds the group order by $M(n) = n \prod_{i=0}^{\lfloor \log_2 n \rfloor - 1} (n - 2^i)$. For $26 \le n \le 63$, exact integer arithmetic in the certificate verifies $M(n) < 2^{n-3}$. For $n \ge 64$, put $k = \lfloor \log_2 n \rfloor \ge 6$. Then

$$\log_2 M(n) < (k+1) \log_2 n < (k+1)^2 \le 2^k - 3 \le n - 3.$$

The middle inequality follows at $k = 6$ and is preserved when k increases by one. Since C-rank is an integer and is less than $n - 3$, it is at most $n - 4$. We have proved the full primitive-parabolic bound for all degrees, conditional only on the explicitly retained Whiston and Maróti theorems and the bounded exact catalogue inputs.

11 A uniform simple-kernel bound

Lemma 10. *Let S be a finite nonabelian simple group. Every subdirect subgroup of S^m is a product of diagonal copies of S on a partition of the m coordinates, with automorphisms allowed in the diagonal maps. A strictly increasing chain of subdirect subgroups has at most $m - 1$ strict steps.*

Proof. Proceed by induction on m. Project onto the first $m - 1$ coordinates. The kernel in the last factor is normal in S, so is either S or trivial. In the first case the subgroup is the product of its projection with S. In the second it is the graph of an onto homomorphism from that projection to S. By induction the projection is a direct product of simple diagonal factors. Their images are commuting normal subgroups of S, so exactly one has nontrivial image, and its map is an isomorphism. The last coordinate therefore joins that diagonal block. This proves the description. In an inclusion of such subgroups the larger group can only refine the partition. If the number of blocks does not increase, their orders are equal and the groups coincide. There are between one and m blocks, proving the chain bound. □

This is the usual subdirect-product lemma, often called Scott's lemma; its complete elementary proof is included here, rather than retained as an unpriced extra premise.

Lemma 11. *Suppose $N \trianglelefteq G$ is a subdirect subgroup of S^m for a finite nonabelian simple group S, and conjugation by G permutes the coordinates transitively, using automorphisms on individual factors. Then*

$$\mu(G) \le \mu(G/N) + \mu'(S) + m - 1. \tag{9}$$

Proof. Normalize an irredundant generating family as in Lemma 3, obtaining $T \cup E$, with T minimally surjecting onto G/N and $E \subset N$. The normalized extra elements stay independent relative to $\langle T \rangle$, by the substitution argument, and no member of T can be dropped since its quotient image is indispensable. Thus this replacement preserves irredundance. Write $T_0 = \langle T \rangle$ and $N_0 = T_0 \cap N$. The subgroup generated by all T_0-conjugates of E is normalized by T_0 and contains E, hence is normal in G. It follows that N is generated by N_0 and these conjugates.

Choose a minimal subset $B \subseteq E$ such that the projections of N_0 and of all T_0-conjugates of B generate S in the first coordinate. Then $|B| \leq \mu'(S)$. To see this, for each $b \in B$ choose an element in the projected subgroup generated by its conjugates which does not belong to the subgroup generated by N_0 and the corresponding subgroups for the other members of B. These selected elements form an independent family in S.

Put $J = \langle T, B \rangle$. The group $J \cap N$ projects onto the first copy of S. Conjugation by $N \leq S^m$ fixes each coordinate, so the transitive coordinate action of G is already that of T_0. Therefore it projects onto every copy. Add the elements of $E \setminus B$ one by one. Each addition strictly enlarges the generated group, by irredundance. Since the image modulo N stays fixed, it strictly enlarges its intersection with N. These intersections form a chain of subdirect subgroups, with at most $m-1$ strict steps by Lemma 10. Thus $|T| + |E| \leq \mu(G/N) + \mu'(S) + m - 1$, as required. □

12 Primitive block stabilizers

Lemma 12. *Let L be primitive of degree $k \geq 5$, with $L \neq A_k, S_k$. Then $\mu'(L) \leq k - 3$, except for the degree-six copy of S_5, which has $\mu' = 4$.*

Proof. For $k \geq 26$, the Maróti and Whiston argument above bounds all independent families, not only C-group markings, by $k - 4$. For $5 \leq k \leq 25$, the complete primitive catalogue and the prime-factor bound leave exactly the following eleven entries:

5	$C_5 : C_4$
6	$A_5,\ S_5$
7	$PSL_3(2)$
8	$AGL_3(2),\ PGL_2(7)$
9	$AGL_2(3),\ PSL_2(8) : 3$
10	$A_6 : 2^2$
12	M_{12}
16	$AGL_4(2)$.

(10)

The same required program checks this complete list and the exact maximal-subgroup, normal-subgroup and quotient facts below.

The group $C_5 : C_4$ has irredundant length at most two by its minimal normal C_5 and cyclic prime-power quotient. Every proper subgroup is cyclic or dihedral of order ten, hence also has bound two. Whiston treats A_5 and S_5 in their natural degrees five. Every maximal subgroup of $PSL_3(2)$ is S_4 or has at most two prime factors, so a generating family has length at most four and a proper subgroup has independent length at most three. The index-two extension $PGL_2(7)$ consequently has independent length at most five. Here and below we use the immediate subgroup version of Lemma 3: $\mu'(G) \leq \mu'(G/N) + \mu'(N)$.

For $AGL_3(2)$ consider any subgroup H, its intersection U with the translation module V, and its linear image $P \leq GL_3(2)$. If $\dim U = 0$, the bound is four. If $\dim U = 1$ or two, P lies in a line or plane stabilizer, each isomorphic to S_4 (its normal 2^2 and quotient $GL_2(2)$ act as the full affine group on four points), so the bound is $3 + \dim U \leq 5$. If $U = V$ and P is irreducible, the normal module costs one and the quotient costs at most four. If P is nontrivial and reducible, it lies in an S_4 stabilizer, and the module costs at most two: three independent vectors would, by Lemma 6, identify V with a sum

of three one-dimensional, hence trivial, modules, contradicting the faithful nontrivial action. Finally $P = 1$ costs only three. Thus every H has irredundant length at most five.

Every proper subgroup of $GL_2(3)$ lies in a maximal subgroup whose order has at most four prime factors; the whole group has irredundant length three by its Frattini quotient S_4. Thus $\mu'(GL_2(3)) \le 4$, and $\mu'(AGL_2(3)) \le 2 + 4 = 6$. The earlier maximal-subgroup bound for $PSL_2(8)$ gives independent length at most five, so its index-three extension has bound six. The degree-ten group has normal A_6 and quotient of order four, giving at most $4+2 = 6$. Every maximal subgroup of M_{12} has at most eight prime factors, so its independent length is at most nine. Finally $AGL_4(2)$ has a four-dimensional normal translation module and quotient A_8, giving $\mu' \le 4 + 6 = 10$. All satisfy the asserted bounds. □

13 The full imprimitive bound

Theorem 13. *Every transitive imprimitive permutation group of degree $n \ge 9$ has maximum irredundant generating length at most $n - 4$.*

Proof. Choose a minimal nontrivial block, of size k, and let $m \ge 2$ be the number of its translates, so $n = km$. The induced group L of the setwise block stabilizer is primitive: a proper block for L would propagate to a smaller block system for G. After choosing identifications of the blocks by elements of G, the group embeds in $L \wr S_m$. Its block image P is transitive, and the projection K_1 of the block kernel K is normal in L; all other projections are conjugate to it.

Blocks of size two, three or four are covered by Theorems 5, 8 and Lemma 9. Assume $k \ge 5$. If $m \ge 3$ and $L < S_k$, then every subgroup of L has independent length at most $k - 2$, by Whiston's equality clause. Successive coordinate projections give $\mu'(K) \le m(k - 2)$, and therefore

$$\mu(G) \le (m - 1) + m(k - 2) = n - m - 1 \le n - 4.$$

If $m = 2$ and $L \ne A_k, S_k$ is not the degree-six copy of S_5, Lemma 12 similarly gives $\mu(G) \le 1 + 2(k - 3) = n - 5$.

It remains to treat $L = A_k, S_k$, and the exceptional degree-six copy of S_5 with $m = 2$. First take $L = A_k$ or S_k. By simplicity of A_k and its trivial centralizer in S_k, the normal subgroup K_1 is either trivial, A_k, or S_k. In the trivial case $K = 1$ and $\mu(G) \le m - 1 \le n - 4$. Otherwise $N = K \cap A_k^m$ is subdirect in A_k^m: the commutator subgroup of K already projects onto A_k in every coordinate. The quotient G/N embeds in $C_2 \wr S_m$. Its maximum irredundant length is at most $(m - 1) + m = 2m - 1$ by the normal-module bound, so Lemma 11 and $\mu'(A_k) \le k - 2$ give

$$\mu(G) \le (2m - 1) + (k - 2) + (m - 1) = k + 3m - 4.$$

For $k \ge 5, m \ge 3$ this is at most $km - 4$, since $(k - 3)(m - 1) \ge 3$. When $m = 2$, the quotient lies in $C_2 \wr S_2 \cong D_8$; every subgroup of D_8 has irredundant length at most two. The sharper bound is then $\mu(G) \le 2 + (k - 2) + 1 = k + 1 \le 2k - 4$.

Finally let $k = 6, m = 2, L \cong S_5$. Use the characteristic subgroup $A_5 \lhd L$ and the same argument in the abstract factors. The block kernel is trivial or projects onto A_5 or S_5. In the latter two cases Lemma 11, now with $S = A_5$, gives $\mu(G) \le 2 + 3 + 1 = 6 \le 12 - 4$. This completes all cases and every parameter value. □

Corollary 14. *Let $(S_n; \rho_0, \dots, \rho_{r-1})$ be a C-group with $n \ge 9$ and $r \in \{n - 2, n - 1\}$. Every maximal standard parabolic is intransitive on the n letters.*

Proof. Each such parabolic has an independent distinguished family of $r - 1 \ge n - 3$ elements. The primitive bound and Theorem 13 exclude every transitive proper subgroup except A_n. Suppose one parabolic is A_n and let a be the omitted involution. For each other distinguished involution b, the intersection property says $A_n \cap \langle a, b\rangle = \langle b\rangle$. Since $aba \in A_n$, this forces $aba = b$. Thus a centralizes all of A_n and $S_n = A_n \times \langle a\rangle$, impossible. The centralizer of A_n in S_n is trivial for $n \ge 4$, as is seen by commuting with all three-cycles. Hence no maximal parabolic is transitive. □

14 The retained extremal inputs

The classification proved in this section is Cameron–Cara, Theorem 2.1, with its strong intersection and relative-subgroup consequences. Its proof is now common to two applications. The ordinary and parity-modified families remain their source results; the present organization makes no new historical classification claim. **W1 (Whiston bound and extremal generation).** An independent family in S_m has at most $m-1$ members, and an independent family with $m-1$ members generates S_m. We use precisely this implication, not a converse asserting that every independent generating family has maximum size.

W2 (Whiston omission consequence). If $n \geq 7$ and (s_i) is an independent generating family of S_n of size $n-1$, each omission subgroup $\langle s_j : j \neq i \rangle$ is intransitive, is transitive imprimitive with blocks of size two, or is A_n. This is the separate consequence of Whiston's proof stated on page 4 of the 2002 manuscript. Its dependence on the classification of finite simple groups is retained. W2 is not deduced here from W1. Cayley's labelled-tree count and the innerness of every automorphism of S_n for $n \geq 7$ are retained classical inputs. The CFSG-dependent omission consequence W2 is separate from W1.

15 Tree markings and the parity construction

Theorem UB.Sign (sign-kernel extension). Let the blocks be singletons and distinguish an edge d, with $s = \tau_d$. For $e \neq d$, choose $\epsilon_e \in \{1, -1\}$ and put

$$r_d = s, \qquad r_e = (s\tau_e)^{\epsilon_e}. \tag{11}$$

For $U \subseteq E$, set $F(U) = \{d\} \cup (U \setminus \{d\})$. Then

$$\langle r_e : e \in U \rangle = \begin{cases} K_{F(U)}, & d \in U, \\ K_{F(U)} \cap A_\Omega, & d \notin U. \end{cases} \tag{12}$$

Both the ordinary marking (τ_e) and (11) generate S_Ω, are independent, and satisfy the full subset intersection property with $B = 1$.

Proof. Inverting an individual generator leaves its cyclic subgroup unchanged. If s is included, s and $s\tau_e$ recover τ_e, proving the first case. The sign kernel of K_F, for $d \in F$, is generated by the $s\tau_e$, $e \in F \setminus \{d\}$: every even word in edge transpositions is a product of pairs, and

$$\tau_e\tau_f = (s\tau_e)^{-1}(s\tau_f), \qquad s\tau_d = 1.$$

This proves the second case, including the empty set, since $K_{\{d\}} \cap A_\Omega = 1$. Intersect (12) using Theorem UB.F and the sign condition. Since $F(U) \cap F(V) = F(U \cap V)$, the result is exactly the subgroup for $U \cap V$. The same conclusion for ordinary markings follows from the unit-weight specialization of Theorem UB.F. Omitting an ordinary edge or a twisted label $e \neq d$ leaves its cut invariant; the omitted generator crosses that cut. Omitting r_d leaves only even permutations, whereas r_d is odd. These observations prove independence and the generation assertions. ■

16 The extremal converse

Corollary UB.Blocks (the two-point-block obstruction). If $m \geq 4$ and $H \leq C_2 \wr S_m$ is transitive on $2m$ points, then H has no independent generating family of size $2m-2$.

Proof. Let N be the block kernel and P the transitive block image. If $P < S_m$, Whiston's equality clause gives $\mu(P) \leq m-2$, and Lemmas 3 and 7 give $\mu(H) \leq 2m-4$. If $P = S_m$, their stronger module bound gives $\mu(H) \leq m+1$. Both quantities are strictly smaller than $2m-2$ for $m \geq 4$, including the endpoint $m = 4$. □

Theorem UB.Extremal (extremal recognition). Assume $n \geq 7$. Every independent family of $n-1$ elements of S_n is an ordinary unit-weight tree marking from Theorem UB.F or a sign-kernel marking (11). If the family is independent relative to $B \leq S_n$, then $B = 1$.

Proof. By W1 the family generates S_n. For its omission groups G_i, W2 and Corollary UB.Blocks exclude the two-point-block case: in that case $n = 2m \geq 8$, and the remaining $n-2$ generators are an independent generating family for G_i. At most one G_i is A_n, because its omitted generator must be the unique odd member. Label a possible exception by 1.

For each $i > 1$, choose an edge $e_i = \{x, s_i(x)\}$ crossing two G_i-orbits. Such an edge exists, since otherwise every generator would preserve those orbits. Choose e_1 from any moved point of s_1. If all the omission groups are intransitive, choose e_1 crossing its own omission orbits as well. For $j \neq i$, the edge e_j lies inside a G_i-orbit because $s_j \in G_i$. In particular the chosen edges are distinct: a repeated edge with a label $i > 1$ would both cross and stay inside the G_i-orbits. Thus a cycle cannot contain e_i for $i > 1$: the remaining edges would give a path within a G_i-orbit between its endpoints. A cycle cannot consist of e_1 alone. The $n-1$ edges therefore form a tree on the n points.

Deleting e_i, for $i > 1$, leaves two connected sets each contained in a G_i-orbit; intransitivity makes them exactly its two orbits. Hence s_j preserves this cut whenever $j \neq i$. Intersect these cut stabilizers using Theorem UB.F. It follows that $s_1 \in K_{\{e_1\}}$, so $s_1 = \tau_{e_1}$, and that $s_i \in K_{\{e_1,e_i\}}$. If every omission group is intransitive, include the first cut too to obtain $s_i = \tau_{e_i}$ for every i. Otherwise $G_1 = A_n$, and all other generators are nonidentity even elements of the two-edge group. For adjacent edges that group is S_3, whose nonidentity even elements are the two inverse three-cycles; for disjoint edges it is $C_2 \times C_2$, with a unique nonidentity even element. These are precisely $(s_1\tau_{e_i})^{\pm 1}$, giving (11).

Now suppose the original family is independent relative to B. It is independent in the ordinary sense, so the preceding classification applies. By Theorems UB.F and UB.Sign its omission groups are the tree-cut stabilizers, except that in the sign-kernel case the distinguished omission group is A_n. At each maximal omission group independence forces $B \leq G_i$; otherwise $\langle B, G_i \rangle = S_n$. A pendant edge different from the distinguished edge exists and yields a point stabilizer, so B fixes a point. At a nonmaximal cut, Lemma UB.Overgroups forces B into the unordered two-block stabilizer. Its elements outside the ordered stabilizer interchange the halves and fix no point. Thus $B \leq G_i$ here also. Strong independence from Theorems UB.F and UB.Sign gives $B \leq \bigcap_i G_i = 1$. ■

17 A finite recognition interface

For a partition π of a finite set Q, write $\Upsilon(\pi) = \prod_{B \in \pi} \mathrm{Sym}(B)$, as a subgroup of $\mathrm{Sym}(Q)$. Let $(h_i)_{i \in I}$ be a finite family in $\mathrm{Sym}(Q)$ and let $H_S = \langle h_i : i \in S \rangle$ for $S \subseteq I$. Fix $d \in Q$. Suppose each S is assigned a partition π_S of Q. If $x \notin Q$, let π_S^x be the partition of $Q \cup \{x\}$ obtained by adjoining x to the block containing d.

We impose the following finite conditions, for every $S, R \subseteq I$:

$$H_S \cap H_R = H_{S \cap R}, \qquad H_S \cap \Upsilon(\pi_R) = H_{S \cap R}, \tag{13}$$

$$\pi_S \wedge \pi_R = \pi_{S \cap R}, \qquad \langle H_S, (d\ x) \rangle = \Upsilon(\pi_S^x). \tag{14}$$

Here the meet is common refinement. In particular, taking $S = R$ in (13) gives $H_S \leq \Upsilon(\pi_S)$. These hypotheses concern subgroups on $|Q|$ or $|Q|+1$ letters; they do not assume the desired intersection property on any larger degree.

Let T be a finite tree meeting Q in exactly $\{d\}$, with at least one edge, and suppose d is a leaf of T. Put $\Omega = Q \cup V(T)$. Alongside the h_i, take the transposition t_e exchanging the endpoints of each edge e of T. All these permutations fix the unspecified letters.

Theorem 15. *Under* (13)–(14)*, the family* $(h_i)_{i \in I}, (t_e)_{e \in E(T)}$ *has the full subset intersection property. If the* h_i *are distinct nonidentity involutions, then so is this enlarged family and it is a C-group marking of its generated group. If moreover* $\pi_I = \{Q\}$*, that group is* $\mathrm{Sym}(\Omega)$*.*

Proof. For $F \subseteq E(T)$, let D_F be the vertex set of the component containing d in $(V(T), F)$; the other component sets will be denoted C. The tree transpositions on a connected component generate its full symmetric group: along the unique paths, conjugating endpoint transpositions produces every transposition from one fixed vertex. Thus

$$\langle h_i \ (i \in S), t_e \ (e \in F) \rangle = H_S[D_F] \times \prod_{C \neq D_F} \mathrm{Sym}(C), \tag{15}$$

where $H_S[\{d\}] = H_S$. If $D_F \neq \{d\}$, then $H_S[D_F]$ is the Young subgroup obtained from π_S by adjoining $D_F \setminus \{d\}$ to its d-block. Indeed, choose any $x \in D_F \setminus \{d\}$. The subgroup $\mathrm{Sym}(D_F)$ contains $(d\ x)$, so (14) supplies every transposition within each unchanged core block and within the enlarged d-block on $Q \cup \{x\}$. Conjugation by $\mathrm{Sym}(D_F)$ supplies the same transpositions for every point of D_F. Conversely all the original generators preserve the claimed blocks. This proves (15).

We now intersect the groups for (S, F) and (R, E). The partition calculation in UB.F applies to their forest components: common refinement is the component partition of $F \cap E$. The remaining issue is the core factor, which need not itself be a Young subgroup.

If $D_F = D_E = \{d\}$, the core intersection is $H_{S \cap R}$ by (13); all other factors follow from the forest argument. Suppose exactly one of D_F, D_E is a singleton, say $D_F = \{d\}$. Every permutation in the first group preserves Q setwise and acts there in H_S. Its additional membership in the second group restricts this action exactly to $\Upsilon(\pi_R)$. Equation (13) gives $H_{S \cap R}$ on Q. Off Q, the two forest component partitions have the refinement just described; their intersections with the enlarged root component are connected subtrees too. Consequently the remaining factors are exactly the symmetric groups on the nonroot components of $F \cap E$.

Finally suppose both root components are nonsingletons. The unique edge incident with d lies in both F and E. Therefore $D_{F \cap E} = D_F \cap D_E$ is also nonsingleton. Both groups in (15) are now Young subgroups of $\mathrm{Sym}(\Omega)$. Their intersection is the Young subgroup of the common refinement. On the core this is $\pi_S \wedge \pi_R = \pi_{S \cap R}$; the root block acquires exactly $D_F \cap D_E \setminus \{d\}$, and all other blocks are the nonroot components of $F \cap E$. This is precisely (15) for $(S \cap R, F \cap E)$. All subset intersections have been proved.

No tree transposition is supported on Q, and different edges give different transpositions. Thus the distinctness assertion follows. For the whole edge set, $D_F = V(T)$ is nonsingleton; if π_I has one block, (15) is $\mathrm{Sym}(\Omega)$. □

18 A uniform presentation for a tree

Let T be a finite tree on n vertices. Write s_e for its edge transpositions. The following is the presentation in Fernandes–Leemans, Proposition 3.1, without its unnecessary restriction $n \geq 7$.

Theorem 16. *The symmetric group on the vertices of T has generators s_e and relations*

$$s_e^2 = 1, \qquad (s_e s_f)^2 = 1 \quad (e, f \textit{ disjoint}), \qquad (s_e s_f)^3 = 1 \quad (e, f \textit{ meet}),$$

together with $(s_e s_f s_e s_g)^2 = 1$ for every three distinct edges e, f, g incident with one vertex. For each unordered triple it is enough to impose the relation in one ordering; all the other orderings then follow.

Proof. First observe that the choice of ordering is immaterial. For three incident involutions a, f, g, the pairwise braid relations give

$$[afa, g] = 1 \iff [a, fgf] = 1 \iff [a, gfg] = 1 \iff [aga, f] = 1.$$

The first and last equivalences follow by conjugation and a braid identity. Also $afa = faf$ interchanges the first two letters. These interchanges generate all orderings of the triple.

Let $P(T)$ be the presented group. The indicated transpositions satisfy the relations and generate $\mathrm{Sym}(V(T))$, so there is a surjection $P(T) \to \mathrm{Sym}(V(T))$. We prove $|P(T)| \le n!$ by induction on n. The cases $n \le 2$ are immediate. Remove a leaf p, with neighbor q, and write $a = s_{pq}$. By induction the subgroup H generated by the remaining edge generators is a quotient of $\mathrm{Sym}(V(T)\setminus\{p\})$. Its displayed permutation image is that entire symmetric group, so the quotient is faithful and we may identify H with S_{n-1}.

We first prove that a centralizes H_q, the point stabilizer of q in H. It commutes with every edge not incident with q. If $f = s_{qu}$ and $g = s_{qv}$ are two distinct other edges at q, the triangle relation says that afa commutes with g. The braid identity $afa = faf$ turns this into the assertion that a commutes with $fgf = (uv)$. The edge transpositions within the components of $T - \{p, q\}$, together with these transpositions between their neighbors at q, form a connected transposition graph on all vertices other than p, q. They therefore generate H_q. This proves the assertion, including the case where there is at most one such component.

Choose a neighbor u of q in $T - p$. The relation for a and s_{qu} is the braid relation. Conjugating it by H_q, which acts transitively on the remaining $n - 2$ vertices and centralizes a, gives

$$af_v a = f_v a f_v \qquad \text{for } f_v = (qv), \quad v \ne p, q.$$

Set $f_q = 1$. Every element of H has a unique left coset of H_q, represented by some f_v. Consequently the union of the following at most n left cosets of H is stable under right multiplication by H:

$$H, \qquad Haf_v \quad (v \ne p).$$

Explicitly, $f_v h = k f_w$ for some $k \in H_q$, so $Haf_v h = Hakf_w = Haf_w$. It is stable under multiplication by a as well: $Ha \cdot a = H$, and for $v \ne q$ the braid identity gives $Haf_v a = Hf_v af_v = Haf_v$. The union contains the identity, and $P(T)$ is generated by H and a. It is therefore all of $P(T)$, whence $[P(T) : H] \le n$ and $|P(T)| \le n!$. The surjection to S_n is an isomorphism. □

19 Transporting an exact presentation

Lemma 17. *Suppose $K = \langle s_e \mid R(s)\rangle$. Let $r_i = v_i(s)$ generate K and choose words $w_e(r)$ with $w_e(v(s)) = s_e$ in K. Then an exact presentation on the marking r_i is*

$$\langle r_i \mid R(w(r)), \quad r_i = v_i(w(r)) \text{ for every } i\rangle. \tag{16}$$

Proof. The prescribed elements of K satisfy all displayed relations, giving a surjective map from (16) to K. Conversely $s_e \mapsto w_e(r)$ defines a homomorphism from K to (16), because every relator $R(w)$ is imposed. One composite fixes each s_e by the word identity in K; the other fixes each r_i by the second group of relations. Thus these homomorphisms are inverse. □

20 Attribution and formal limits

The existing source attributions remain in each proof and in the five applications. The common extremal classification retains Whiston's independent-family theorem and omission consequence separately. Maróti's primitive-group theorem and the exact finite GAP catalogue are retained in the all-degree bound. The entire primitive-bound program is required for its bounded exceptions; the three other programs retain their validation and auxiliary-example roles. All four files remain complete in the program appendix.

The existing ResidualGeneration and IrredundantNormalization Lean files are retained unchanged. They verify the subgroup induction and the actual normalization/independence leg, respectively. They do not formalize laminar completion, the classification inputs, the core attachment theorem, finite catalogue completeness, the presentations or any full source geometry. No additional formal declaration is claimed by this reorganization. All full costs of this common paper and the mathematical program bundle belong to this topic.

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